



Review paper

Landscape connectivity and insect herbivory: A framework for understanding tradeoffs among ecosystem services



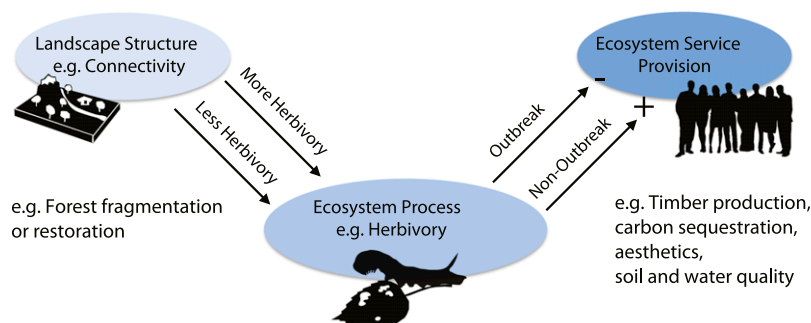
Dorothy Y. Maguire^{a,*}, Patrick M.A. James^b, Christopher M. Buddle^a,
Elena M. Bennett^{a,c}

^a McGill University, Department of Natural Resource Sciences, Canada

^b Université de Montréal, Département de Sciences Biologiques, Canada

^c McGill School of the Environment, Canada

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 7 April 2015

Received in revised form 18 May 2015

Accepted 19 May 2015

Available online 27 May 2015

Keywords:

Connectivity
Insect herbivory
Ecosystem services
Outbreking insects
Ecosystem processes
Forests

ABSTRACT

Current theory suggests that ecosystem services in fragmented landscapes can be maintained by preserving connectivity of remaining habitat patches. However connectivity does not always influence services positively. For example, outbreaks of destructive insect herbivores can be facilitated by connectivity among forest patches. Understanding the positive and negative effects of connectivity on ecosystem processes is needed to help scientists and managers anticipate tradeoffs among services that result from forest fragmentation or restoration. In this paper we use a vote counting meta-analytic approach in combination with a literature survey to explore how connectivity affects ecosystem service provisioning using insect herbivory as a model process. Our results indicate that landscape connectivity affects herbivory in diverse ways, and that implications for services depend on whether we consider outbreaking species. Under non-outbreak conditions, herbivory positively affects services such as timber production, soil formation, and recreation by stimulating tree growth and enhancing soil productivity, but under outbreak conditions, herbivory neg-

* Correspondence to: USDA-ARS European Biological Control Laboratory, 810 Avenue du Campus Agropolis, Montferrier sur Lez, 34980, France.

Tel.: +33 (0) 4 99 62 30 05; fax: +33 (0) 4 99 62 30 49.

E-mail address: maguire.dory@gmail.com (D.Y. Maguire).

<http://dx.doi.org/10.1016/j.gecco.2015.05.006>

2351-9894/© 2015 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

actively affects services by reducing timber yields and the aesthetic value of forests. We present a framework that shows herbivory is an important mechanism through which connectivity affects ecosystem services. Using case studies we demonstrate the applicability of the framework to management of two forest insect pests: the mountain pine beetle and forest tent caterpillar.

© 2015 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Contents

1. Introduction.....	74
1.1. Insect herbivory	75
2. Literature survey methods	75
2.1. The effects of connectivity on herbivory	76
2.2. The effects of herbivory on ecosystem services.....	76
2.3. Framework	76
3. Results and discussion	76
3.1. Connectivity and herbivory	76
3.1.1. Ecosystem type	77
3.1.2. Population dynamics	77
3.2. Herbivory and ecosystem services	78
3.2.1. Provisioning service: timber production	78
3.2.2. Cultural service: aesthetic value of forests	78
3.2.3. Supporting service: soil quality	79
3.2.4. Regulating service: carbon sequestration	79
3.2.5. Summary of the relationships between herbivory and ES.....	79
3.3. Framework	79
3.4. Case studies	80
3.4.1. Case study 1: negative effects of connectivity on services caused by herbivory	80
3.4.2. Case study 2: positive effects of connectivity and services affected by herbivory	80
4. Conclusions.....	81
Acknowledgements.....	82
Appendix A. Supplementary data	82
References.....	82

1. Introduction

Human activities such as land clearing for agriculture alter the composition and configuration of natural areas and affect their ability to provide ecosystem services (ES) (Foley et al., 2005; Vitousek et al., 1997). Human activities are generally focused on increasing provisioning ES such as food or timber, often at the expense of cultural and regulating ES such as recreation or climate regulation (Tilman et al., 2011). Increasing landscape level connectivity has been proposed as one way to mitigate the negative effects of human activities on ES (Calabrese and Fagan, 2004; Fischer and Lindenmayer, 2007; Taylor et al., 1993). However, changes in connectivity, whether decreasing as a result of clearing for agriculture and urban expansion, or increasing as a result of restoration and conservation, may have both positive and negative effects on ES through complex interactions with different species and processes (Mitchell et al., 2013). Careful evaluation of the effects of landscape connectivity on multiple ES are needed to identify tradeoffs that result from changes in connectivity.

Increased connectivity of habitat may enhance pollination by facilitating pollinator dispersal (Kremen et al., 2007). Connectivity can also enhance the recreational value of forests by facilitating the movement and persistence of wildlife (Coulon et al., 2004) and the movement of hunters and hikers (Vanderzee, 1990). Meta-community theory also suggests that connectivity is important for maintaining biodiversity (Leibold et al., 2004), which according to the spatial-insurance hypothesis (Leibold et al., 2004; Loreau et al., 2001, 2003) may enhance ecosystem function and improve ES indirectly (Balvanera et al., 2006). For example, carbon sequestration may be positively affected by connectivity through its indirect effects on tree species diversity (Ziter et al., 2013). Conversely, connectivity may also have detrimental effects on ES. Increased spatial connectivity is known to facilitate the establishment and spread of diseases (Biek and Real, 2010; Blanchong et al., 2008) and invasive species (Crowl et al., 2008). Connectivity in agricultural systems may facilitate the spread of crop pests (Margosian et al., 2009). Occasionally, both positive and negative effects can be observed at the same time: in stream systems connectivity enhances ES such as water quality by facilitating nutrient cycling and allowing the passage of water filtering species such as mussels (Pringle, 2003). This same connectivity can degrade ES by facilitating the passage of invasive or noxious species (Pringle, 2003).

There is little theoretical basis for understanding and predicting the consequences of changes in connectivity on ES. Understanding when to expect increases, and when to prepare for decreases in ecosystem service provisioning can help us better assess the consequences of different land use management strategies on multiple, interacting ES.

1.1. Insect herbivory

Herbivory (see Glossary for definitions) is a dynamic process that interacts with many other processes (Yang and Gratton, 2014), influences ES both negatively and positively (Schowalter, 2012), and is affected by landscape connectivity (Thies et al., 2003; Tscharrntke et al., 2002). These attributes make this process good model for studying how connectivity affects ES. Although herbivory by vertebrates is important in many systems (e.g. Bryant et al., 1983; Ford and Grace, 1998 and van der Wal et al., 2004), here we focus on herbivory by insects because less is understood of how the population dynamics of insect herbivores (e.g. Myers and Cory, 2013) affect ES, and because insect herbivory has ubiquitous, and diverse relationships with ES (Schowalter, 2012).

Insect herbivory (i.e. any transfer of energy from plants to insects through feeding) is a key ecosystem process that influences many ES. For example, by influencing primary productivity (McNaughton et al., 1989) and modifying tree growth, survival, and reproduction (Hochwender et al., 2003) insect herbivory can affect ES such as timber production (Pedigo et al., 1986; Schowalter, 2012), and carbon sequestration (Dymond et al., 2010; Kurz et al., 2008). By influencing tree species recruitment dynamics (Hochwender et al., 2003), modifying the demographics and succession of forests (Barbosa et al., 2005), and enhancing forest biodiversity (Rinker and Lowman, 2004), herbivory affects the recreational value of forests (Sheppard and Picard, 2006). Finally, by altering nutrient dynamics (Belovsky and Slade, 2000; Frost and Hunter, 2004; le Mellec et al., 2011), insect herbivory can affect soil (Hunter, 2001) and water quality (Mikkelsen et al., 2013). There is little synthesis of the literature available on the effects of herbivory on ES, and most of the literature that exists examines high or outbreaking levels of herbivory only (except see Schowalter (2012)). This leaves a critical gap in research as herbivory is a dynamic process that fluctuates spatially and temporally, and with the population dynamics of species.

The relationship between insect herbivores and ES depends on the amount of damage they inflict, which is governed in part by their population dynamics. Outbreak species may be native or introduced (Kenis et al., 2009) and have a disproportionately negative impact on ES compared to others due to their irruptive population dynamics and ability to inflict high levels of damage (Liebhold, 2012; Wallner, 1987). These species may oscillate from periods of low population density and defoliation or 'latent' periods to periods of high population density and defoliation or 'outbreaks'. Conversely, non-outbreak species tend to exhibit stable population dynamics at low densities, and therefore inflict less damage on hosts, and affect ES differently than outbreak species. These population dynamics therefore likely affect the way we relate herbivory as a process to ES.

The population dynamics of insect herbivores are highly sensitive to the spatial legacies created by land management (Cappuccino et al., 1998; Kareiva, 1990; Robert et al., 2012) including changes in connectivity (Kruess, 2003; Thomas et al., 2001). Connectivity affects herbivory by altering the movement and interactions among insect herbivores, their natural enemies, and host plants (Tscharrntke and Brandl, 2004). For example, less connected landscapes can result in increased herbivory as herbivores are released from control by natural enemies (Kruess and Tscharrntke, 2000; Terborgh, 2001; Roland, 1993). Alternatively, reduced connectivity can reduce herbivory as herbivores become less able to find resources (De La Vega et al., 2012). Under some circumstances, connectivity does not affect herbivory (Souza et al., 2013; Jonsen and Fahrig, 1997). There is no clear consensus emerging from this literature, and the effects of connectivity on herbivory seem to vary widely among species and habitat types.

Although we know that herbivory is affected by changes in landscape connectivity, there are currently no frameworks with which to assess the implications of such changes on ES. For example, little is known about the type of relationship between herbivory and ES, and how this varies with respect to different services. Improved understanding of the relationships among connectivity, herbivory and ES will therefore allow us to evaluate the effects of changes in landscape structure on ES more thoroughly, and to guide future management decisions in a diverse set of ecosystem types.

The purpose of this review and synthesis is to explore how landscape connectivity affects ES provision using insect herbivory as a model system, driven by three interconnected objectives;

- (1) **Connectivity and herbivory:** Determine how landscape connectivity affects insect herbivory and how this effect varies in response to whether the herbivore is an outbreaking or non-outbreaking species, and in different ecosystem types.
- (2) **Herbivory and ecosystem services:** Describe the relationships between herbivory and ES and determine if they vary among provisioning, regulating, supporting and cultural services.
- (3) **Framework:** Organize the results of objectives 1 and 2 into a diagrammatic conceptual framework for the purpose of more easily understanding the effects of connectivity and insect herbivory on ES, and test the efficacy of the framework using case studies.

2. Literature survey methods

To understand the effects of connectivity on herbivory (objective 1) and herbivory on ES (objective 2), we performed two separate detailed literature reviews. The results of these reviews were then compiled into an easy to understand framework (objective 3) in the form of a conceptual diagram. We then demonstrate the efficacy of the framework using 2 case studies.

2.1. The effects of connectivity on herbivory

We conducted a semi-quantitative literature review (e.g. [Mitchell et al., 2013](#)) that included papers that examine the effects of connectivity on herbivory in any terrestrial natural or semi-natural ecosystem type, including forests, woodlands, fields, grasslands, meadows, hedgerows, marshlands and pasture. We used four search engines (Pub Med, ISI Web of Science, Scopus, Google Scholar), and searched for papers available online up to the end of 2014 with the following combinations of words in their title, abstract, or keywords: 'landscape structure', 'landscape connectivity', or 'fragmentation' and 'herbivore(s)', 'herbivory', or 'insect(s)'. Sixty eight of 136 papers were found to be relevant (i.e. examined the response of herbivory or of herbivore populations to increased landscape connectivity), and retained for final analysis.

We determined whether the results of each paper indicated that there was an increase, decrease, no response, or inconclusive responses (e.g., authors detected variability among years, species, etc.) of herbivory or herbivore abundance to increased connectivity. Although there are exceptions, here we assume that population sizes of insect herbivores are roughly correlated with the amount of damage they inflict. We defined connectivity as the degree to which the landscape facilitates movement of species and matter ([Taylor et al., 1993](#); [Fischer and Lindenmayer, 2007](#)), thus we included both structural components, such as composition and configuration of habitat patches, and functional components, such as organism responses to landscape heterogeneity.

Next, we examined whether ecosystem type, or the type of population dynamics of the species, was correlated with the herbivore response to connectivity. After classifying ecosystem type and whether the species was non-outbreak (i.e. has stable population dynamics) or an outbreak species (i.e. has cyclic or eruptive population dynamics), we simply tallied the number of papers in each category using a vote counting methodology (e.g., [Debinski and Holt, 2000](#) and [Mitchell et al., 2013](#)).

2.2. The effects of herbivory on ecosystem services

We used a second literature review to develop predictions of the relationships between herbivory and ES at low and high levels of herbivory. These predictions could allow us to account for variation in feeding levels of insect herbivores (e.g. caused by non-outbreak vs. outbreak species), and/or changes over time (e.g. population cycles of outbreaking insects from latent to outbreak periods).

We focused on forest ecosystems because this is where the effects of insect herbivory on ES has been frequently examined (e.g. [Schowalter, 2012](#)) and because forests are providers of a diverse and important set of ES ([Krieger, 2001](#)). In our review of the literature, we examined four key forest ES: timber production (provisioning), carbon sequestration (regulating), recreation in forests (cultural), and soil production (supporting/ecosystem process). We searched for papers published up to the end of 2014 using all combinations of the following search terms: 'ecosystem service(s)' and 'insect herbivory', 'herbivore(s)', or 'insect(s)'. We then narrowed our search down to target specific ES by searching combinations of 'insect herbivore(s)' or 'insect herbivory' and each service: 'timber production' or 'timber value', 'carbon sequestration' or 'climate regulation', 'soil production' or 'nutrient cycling', 'recreation' or 'recreational value' or 'forest recreation'. We also considered references cited in those relevant papers we found.

For each service category, we determined the level of herbivory each paper measured (low/ high), and assessed the response of the service to each level, and whether the strength of the effects were weak, moderate, or strong. We then determined if the service measured in the study was positively, or negatively affected by the level of herbivory. We used this information to assess the relationships between increasing amounts of herbivory and each ES.

2.3. Framework

We used the results of the two previous literature reviews to build a conceptual framework (in the form of a diagram) summarizing the possible relationships between landscape connectivity and herbivory, and between herbivory and ES. The diagram allows the reader to consider scenarios of changing landscape connectivity, and understand the possible implications for herbivory, and the ES that herbivory may affect. We then explored how it can inform management decisions by applying it to two specific case studies. In the first case study, we use the available literature to explore the effects of connectivity the mountain pine beetle (MPB; *Dendroctonus ponderosae*), a native forest insect pest that is known to respond positively to landscape connectivity ([Raffa et al., 2008](#); [Safranyik et al., 2010](#); [James et al., 2011](#)). In the second case study, we use the available literature to explore the effects of connectivity on forest tent caterpillar (*Malacosoma disstria*) dynamics, another native irruptive species that responds negatively to landscape connectivity ([Cooke and Roland, 2000](#)).

3. Results and discussion

3.1. Connectivity and herbivory

We found no consensus in the literature about the effects of connectivity on insect herbivory. Most studies (33 out of 68 papers) found greater herbivory with increased connectivity ([Fig. 1\(b\)](#)). The remainder of the studies found less (19), no change (12), or inconclusive results (4).

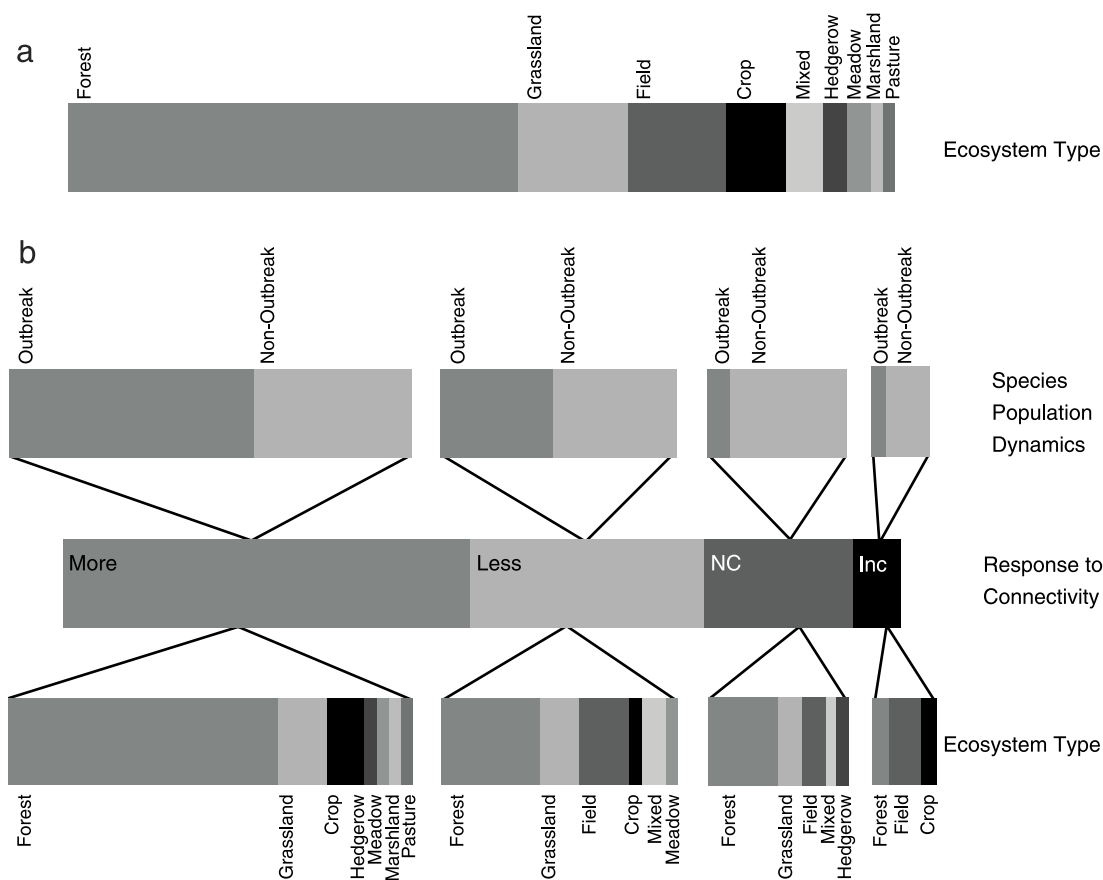


Fig. 1. Summary of the literature survey examining the response of herbivory to increases in landscape connectivity. (a) The proportion of published papers in each ecosystem type. (b) The proportion of published papers that find more herbivory (MORE), less herbivory (LESS), no change in herbivory (NC), or inconsistent (INC) changes in herbivory in response to increased landscape connectivity (centre), the proportion of studies that dealt with species with outbreaking or non-outbreaking population dynamics (top), and the number of studies in each ecosystem type (bottom).

3.1.1. Ecosystem type

The most common landscape types reviewed were forests and woodlands (37 out of 68 papers). Other ecosystems included agricultural areas like fields (8) and croplands (5), and semi-natural areas such as grasslands (9), meadows (2), hedgerows (2), marshlands (1) pastures (1) and mixed ecosystems (3) (Fig. 2(a)). Of the studies that found more herbivory with increased landscape connectivity, most took place in forest ecosystems (67%), and there was a much more even distribution of ecosystem types that found less, no change, or inconclusive results (Fig. 1(b)). These results could suggest that herbivores, at least in forest ecosystems, are more likely to respond positively than negatively to increased landscape connectivity (for example, 60% of forest studies found more herbivory with connectivity than other responses). Studies in other ecosystem types did not show as clear of a trend as those in forests, perhaps because there were fewer studies taking place in other ecosystem types (Fig. 1(b)). These results emphasize the usefulness of forest insects as models for studying land use effects on herbivory, but also encourages the further study of insect herbivores in other ecological systems to better understand the effects of connectivity on herbivory.

3.1.2. Population dynamics

There were a similar proportion of papers that considered species with outbreak and non-outbreak population dynamics in each of the four responses to connectivity (i.e. more, less, no change, or inconsistent) (Fig. 1(b)). These results suggest that landscape connectivity can be used to mitigate the effects of insect outbreaks, but whether to enhance or reduce landscape connectivity will depend on the species, and the context in question. Even studies on the same species (e.g. forest tent caterpillar) in different landscape types (Aspen stands in Canada vs. Maple stands in the USA) have found contrasting results in how this species responds to changes in landscape structure (Roland, 1993; Wood et al., 2010). These results not unsurprisingly draw our attention to the difficulty in prescribing a single management strategy to control insect outbreaks, and suggest that greater focus must be placed on understanding species-specific responses to connectivity. One suggestion is to examine the role of life history characteristics of herbivore species driving responses to connectivity since these are important factors governing the response of species to landscape structure (Ewers and Didham, 2006).

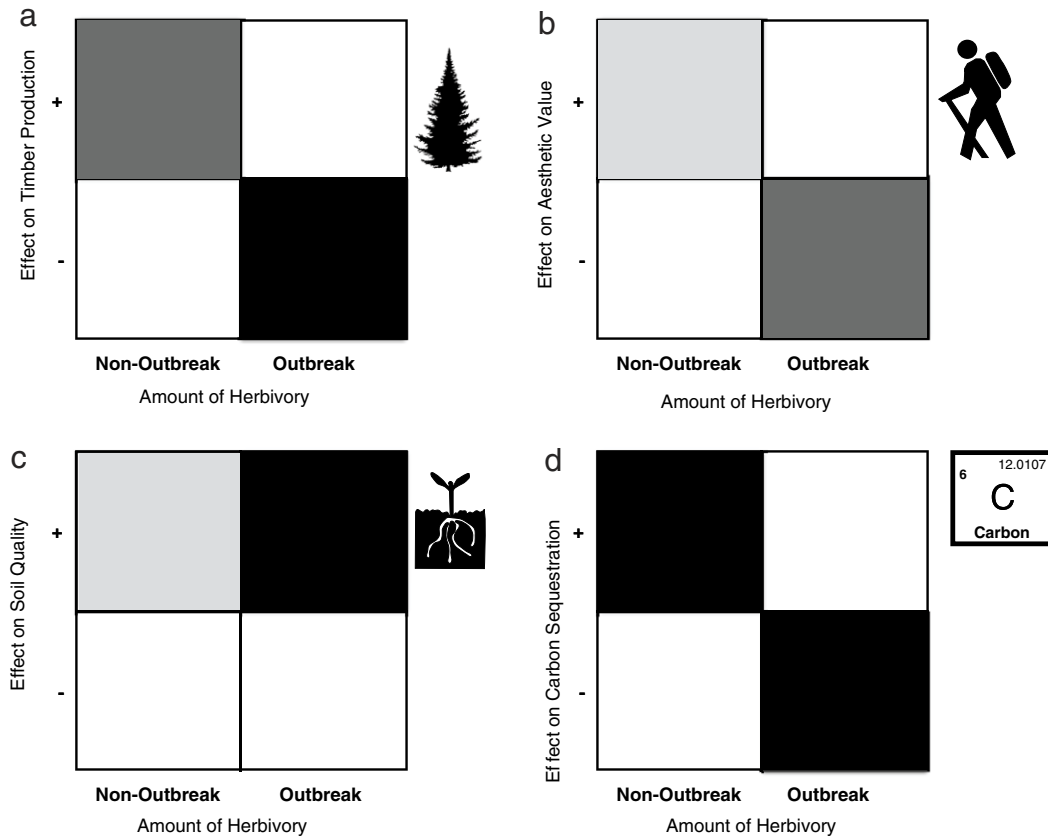


Fig. 2. Visual representations of the hypothesized relationships between insect herbivory and ecosystem services. Specifically (a) timber production, (b) aesthetic value of forests, (c) soil formation and (d) carbon sequestration. Graphs are divided into four sections representing positive and negative effects of herbivory on ES, during non-outbreak (low) vs. outbreak (high) levels of herbivory. Quadrants are coloured differently based on the hypothesized strength of the effect of herbivory on ES: weak (light grey), moderate (dark grey) and strong (black). Proposed relationships are derived from synthesis of the available literature.

3.2. Herbivory and ecosystem services

3.2.1. Provisioning service: timber production

The relationship between herbivory and timber yield (Fig. 2(a)) indicates a moderate positive effect of herbivory on timber at low (non-outbreak) levels of herbivory, and a strong negative effect at high (outbreak) levels. This is based on the observation that initially, damage caused by herbivory either has no effect on growth increment (which affects yield), or a moderate positive effect. The positive effect may occur when plants compensate in response to herbivory (Belsky, 1986), and can result in an increase in timber yield even at larger scales (Dyer et al., 1993; Muzika and Liebhold, 1999; Schowalter and Hargrove, 1986). Beyond some threshold, injury levels begin to have a strong negative effect, where yield significantly decreases with injury level (Pedigo et al., 1986).

Research on insect herbivory and timber production has typically been focused on outbreaking species (e.g., Kulman, 1971 and Schowalter, 2012). While some research suggests low to moderate herbivory can be good for timber yield, there is a paucity of research quantifying these effects, unless it is to determine thresholds for economically efficient pest management (Brandt, 1995; Pedigo et al., 1986). The point at which herbivory negatively affects yield depends on many factors including the tree species (Baraza et al., 2010; Brandt, 1995), the insect involved (Pedigo et al., 1986), site condition and tree “health” (Fuentelba and Bause, 2012; Schowalter and Hargrove, 1986) and the number of years of consecutive defoliation (Schowalter et al., 2011).

3.2.2. Cultural service: aesthetic value of forests

Herbivory affects cultural services such as the aesthetic value of forests (Ribe, 1989) with weak positive effects during low (non-outbreak) levels of herbivory and moderately negative effects during outbreaks (Fig. 2(b)). This is based on the observation that at low and moderate levels, defoliation can have subtle positive effects on forest aesthetics due to increased sunlight and reduced tree density that occurs as a result of herbivory (Sheppard and Picard, 2006; Müller and Job, 2009). However, at higher (outbreak) levels, visual quality may decrease, and the aesthetic value of forests can become

compromised (Hollenhorst et al., 1993) if there is noticeable dead and dying foliage or branches (Buhyoff and Leuschner, 1978) as well as frass deposition. The precise nature of this relationship varies with forest type (Sheppard and Picard, 2006), and the insect feeding style (e.g., defoliator vs. cambium feeder). This relationship may also differ between coniferous or deciduous forest types (Sheppard and Picard, 2006).

3.2.3. Supporting service: soil quality

The relationship between herbivory and soil quality (Fig. 2(c)) suggests that low (non-outbreak) levels of herbivory have positive weak effects on soil quality, that increase during high (outbreak) levels. This is based on the observation that herbivory does not affect soil quality at very low levels (Chapman et al., 2003; Stadler et al., 2001), however, at moderate levels soil quality is improved due to significant inputs of frass, abscised leaves, and bodies to the forest floor (Chapman et al., 2003; Frost and Hunter, 2007; Hunter et al., 2003; le Mellec et al., 2011; Schowalter, 2012). Soil quality continues to improve even after severe or complete defoliation in both the short (Clow et al., 2011) and long term. For example, nutrient inputs after insect outbreaks can continue to have positive effects on future forest growth and productivity up to 30 years post outbreak (Griffin et al., 2011 and references therein). However, these long-term effects may actually be due to the increase in downed wood, and not directly related to herbivory (Griffin et al., 2011). Furthermore, plant and herbivore composition, microclimatic changes as a result of herbivory, underlying site conditions, and other sources of nutrient inputs may influence this relationship (Schowalter and Hargrove, 1986).

3.2.4. Regulating service: carbon sequestration

The relationship between herbivory and carbon sequestration (Fig. 2(d)) shows that low (non-outbreak) levels of herbivory have strong positive effects on carbon sequestration while high (outbreak) levels of herbivory have strong negative effects. This is based on the observation that large increases in carbon sequestration at low to moderate levels of defoliation may be observed because stimulation of plant (DeLucia, 1999; McNaughton et al., 1989), and soil (Metcalf et al., 2013) productivity increase carbon uptake. However, during periods of outbreak, large amounts of defoliation cause significant reductions in tree growth and increased mortality, which convert forests that were carbon sinks, into carbon sources (Dymond et al., 2010; Kurz et al., 2008). Furthermore, recent research shows that increased herbivory in response to elevated CO₂ could reduce the ability of forests to serve as carbon sinks in scenarios of climate change (Couture et al., 2015).

3.2.5. Summary of the relationships between herbivory and ES

We have shown that the relationship between herbivory and ES is unique for each service, and likely non-linear. We also found that the role of herbivory in ES provisioning depends on its intensity, which in turn depends on the population dynamics of the species. For example while non-outbreak species tend to maintain low to moderate levels of feeding, outbreak species will fluctuate from low levels of feeding or “latent” periods to high levels or outbreak periods.

We found that low to moderate levels of feeding, found in non-outbreaking or “latent” periods, seem to provide an overall benefit to ES such as timber production, the aesthetic value of forests, soil quality, and carbon sequestration. Since most of the literature is still focused on negative effects of outbreaks, we believe these results provide incentive for further studies on the benefits of low or moderate levels of herbivory to ES. We find that high levels of feeding, found during outbreaks, are detrimental to most ES (with the exception of soil formation). However, more research is needed to assess the positive and negative short and long-term effects of outbreaks on forest ES (Schowalter, 2012), as well as interactions between outbreaks and other processes and disturbances (Paine et al., 1998) affecting ES.

3.3. Framework

We have summarized the results of objectives 1 and 2 in a conceptual framework (Fig. 3) which illustrates that insect herbivory can be used as a model to understand flows between landscape connectivity and ecosystem services. From the framework, we see that:

- Changes in connectivity may result in more, less or no change in connectivity, depending on the species and system involved.
- Herbivory has unique relationships with different ES that depend on the amount of defoliation (governed by insect population dynamics). While non-outbreak species generally inflict low levels of damage, the level of damage inflicted by outbreak species varies depending on whether it is in the latent or outbreaking phase.
- Changes in landscape connectivity alone will not eliminate outbreaks. Such changes may nonetheless reduce the extent and severity of outbreaks (i.e. create more areas with “latent” levels of herbivory), which will positively affect services. We assume that small fluctuations in low to moderate levels of herbivory as a result of connectivity change will positively affect services if considering non-outbreak species/communities.

As more emphasis is placed on managing landscapes sustainably for the provision of multiple ES (e.g. Bennett et al., 2009; Chan et al., 2006 and Nicholson et al., 2009), it is important that managers be able to anticipate the positive and negative effects of land use decisions (such as those affecting connectivity) on ES. Our framework (Fig. 3) can help managers anticipate the indirect effects of connectivity change on ES affected by herbivory by following these steps:

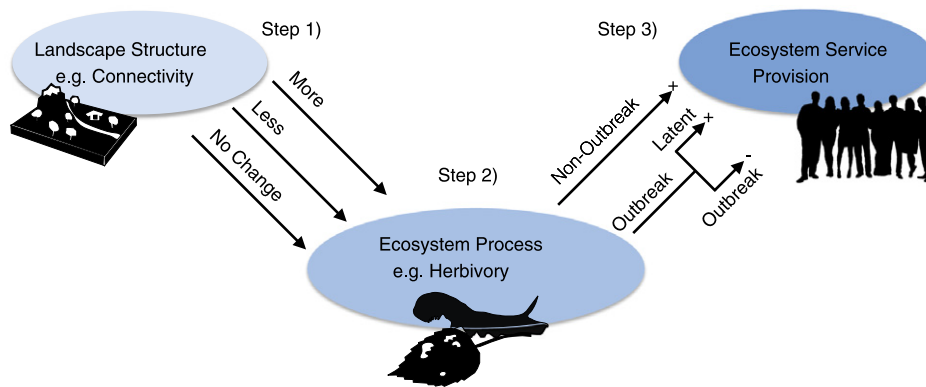


Fig. 3. Conceptual framework illustrating how herbivory acts as a mechanism through which landscape connectivity affects ecosystem service provision. The diagram illustrates the types of relationships (arrows) between connectivity, insect herbivory, and ecosystem services. Whether herbivory has positive or negative effects on services is indicated by positive and negative symbols (+/–). Low to moderate levels of herbivory by non-outbreaking species or outbreaking species in the latent phase, have positive effects on services, while outbreaks have negative effects on services. For a given landscape, one can follow steps 1–3 to help predict how changes in connectivity will alter ES provision via indirect effects on herbivory. Connectivity can either maintain populations of non-outbreak species, or reduce the extent and duration of outbreaks (positively affecting ecosystem services), or facilitate outbreaks (negatively affecting ecosystem services).

- (1) Assess how connectivity might affect herbivory in a given ecosystem. Since the effect of connectivity depends on the species and system involved, managers need to use the available literature to determine if there are species in a region that are likely to inflict more, less, or no feeding damage with changes in landscape connectivity.
- (2) Assess the implications for ES by determining if those herbivore species are outbreak species with the potential to reach high populations and be of economic concern, or whether they are non-outbreak species whose fluctuations in response to connectivity might be too small to make a difference for services. If they are outbreak species, do the changes in connectivity help to maintain areas with latent phase levels of defoliation, or do they facilitate greater extent and severity of outbreaks?
- (3) Based on how herbivore species of interest respond to connectivity, and their population dynamics, one can determine whether management strategies that alter landscape connectivity will have positive or negative effects on ES, and anticipate tradeoffs resulting from the indirect effects of insect herbivores.

3.4. Case studies

3.4.1. Case study 1: negative effects of connectivity on services caused by herbivory

The Mountain Pine Beetle (*Dendroctonus ponderosae*; MPB) is a native outbreaking forest insect pest that is having devastating effects on forests throughout western North America as a result of its recent outbreak. This pest has resulted in billions of dollars in damage and losses to the timber industry (Seidl et al., 2008), it has significantly reduced forest area for carbon sequestration (Kurz et al., 2008), and water quality in the region as well (Uunila et al., 2006). Past management strategies such as fire suppression, and clear cutting have resulted in large homogeneous uninterrupted (connected) tracts of this beetle's host species (pine), which interact with climate warming (Aukema et al., 2008), to produce an unprecedented outbreak (Bone et al., 2013; James et al., 2011; Raffa et al., 2008; Safranyik et al., 2010).

We can use our framework to understand how connectivity, the MPB outbreak, and ES are related in order to better inform future management strategies as follows (illustrated in Fig. 4(a)) (1) We know that management decisions that result in increased connectivity of the MPB's host (pine) is likely going to increase the establishment and spread of this species (Aukema et al., 2008). (2) We know that we are dealing with a native pest species, with outbreaking population dynamics, which will likely negatively affect ES, especially given the recent extent of damage caused by the MPB (Seidl et al., 2008). (3) Taken together we can see that past management focused solely on optimizing timber production (Hennigar et al., 2013) may be convenient in the short term, but the connectivity among MPB host that results may have long term negative feedbacks on ES (e.g. carbon sequestration, recreational value, soil and water quality, ability to support biodiversity) including timber production (Uunila et al., 2006; Kurz et al., 2008). Therefore, using this framework, managers can more explicitly visualize, and anticipate the positive and negative short and long-term effects of changing landscape structure (i.e. connectivity) on forest ES affected by outbreaking insect pests. They can plan against management scenarios that enhance landscape connectivity, and would indirectly facilitate or worsen outbreaks.

3.4.2. Case study 2: positive effects of connectivity and services affected by herbivory

The forest tent caterpillar (*Malacosoma disstria*; FTC) is a native pest, and widespread defoliator of aspen in the boreal forest of North America that outbreaks approximately every decade (Cooke and Lorenzetti, 2006). Unlike the MPB, reduced

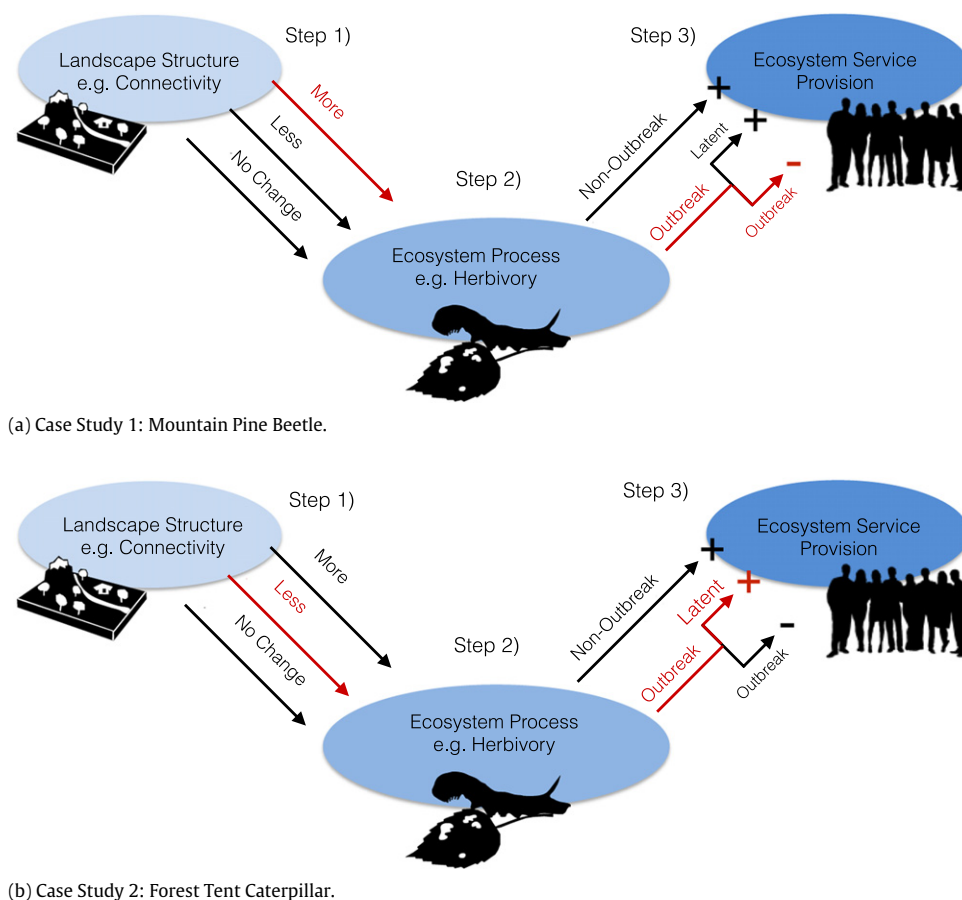


Fig. 4. Diagram of how to apply the framework for (a) case study 1: mountain pine beetle, and (b) case study 2: forest tent caterpillar. The diagram illustrates the types of relationships (arrows) between connectivity, insect herbivory, and ecosystem services. Whether herbivory has positive or negative effects on services is indicated by positive and negative symbols (+/–). Low to moderate levels of herbivory by non-outbreaking species or outbreaking species in the latent phase, have positive effects on services, while outbreaks have negative effects on services. Connectivity can either maintain populations of non-outbreak species, or reduce the extent and duration of outbreaks (positively affecting ecosystem services), or facilitate outbreaks (negatively affecting ecosystem services). Red arrows illustrate the specific relationships emerging from the case studies.

connectivity among host trees as a result of forest fragmentation may benefit this species. The parasitoid species that normally control FTC populations may be negatively affected by fragmentation of its host populations (Roland, 1993; Roland and Taylor, 1997), as they become isolated and difficult to locate (Roland and Taylor, 1997). This disrupts the population dynamics of both the FTC and its natural enemies and releases FTC from predation pressure (Cooke and Roland, 2000), resulting in increased intensity and duration of outbreaks, and lasting consequences for the ES provided by these forests.

Using this framework again, we can see how connectivity, FTC outbreaks, and ES are related in order to better inform future management strategies: (1) We know that increased host connectivity can regulate, or reduce the damage caused by FTC (in certain ecosystems) by facilitating regulation by its natural enemies (Roland and Taylor, 1997; Cooke and Roland, 2000). (2) The FTC is a native pest species that has the potential for periodic outbreaks, which may have negative or positive effects on ES depending on the phase of outbreak (latent or outbreak) (Cooke and Lorenzetti, 2006). (3) Taken together, we see in Fig. 4(b) that maintaining connectivity in agricultural landscapes where FTC is a problem may reduce the extent and severity of outbreaks, resulting in positive effects (or fewer negative effects) on ES affected by herbivory (Cooke and Roland, 2000) such as timber, carbon, aesthetic values of forests, and soil and water quality, as well as other ES that benefit from enhanced connectivity in agricultural landscapes such as pest control, pollination services, and protection against agricultural runoff (Mitchell et al., 2013). This framework can therefore allow managers to visualize even broader synergies, or “win–win” scenarios for multiple ES that might otherwise be missed.

4. Conclusions

Herbivory by insects is an important process to consider in the context of ES and landscape connectivity. Herbivory is sensitive to management decisions at the landscape scale, and has complex relationships with ES. For example, we found that herbivory responds to changes in connectivity in diverse ways (more, less, or no change), and is dependent on the

species and system considered. We have also shown that herbivory has unique relationships with ES that are determined in part by whether the species are outbreaking or non-outbreaking. Our review emphasizes the importance of considering herbivory as an ecosystem process that can mediate the effects of land use change on ES, and thanks to the framework developed in this paper, it is now possible to do that. This information can help land managers think more broadly about the costs and benefits of landscape management on the provision of multiple ES. Finally, we believe this framework, and the methods used to attain it, can be used more generally as a launching point for future studies of the effects of connectivity and other ecosystem processes on ES. By providing a means to weigh the positive and negative effects of landscape connectivity on herbivory, and the corresponding positive and negative effects on ES, this information can be used to better manage landscapes for the provision of a diverse and sustainable set of ES.

Acknowledgements

This research was supported by a McGill University Tomlinson Fellowship in Forest Ecology (DYM) (00078), an NSERC Strategic Projects Grant (EMB), and NSERC Discovery Grants (PMAJ, CMB (RGPIN 261399-2013), EMB (RGPIN 327077-2013)).

Glossary

Ecosystem services: The benefits people obtain from ecosystems that support survival and quality of life. Ecosystem services are supported by ecosystem processes. Ecosystem services can be provisioning (e.g. food, fibre, medicinal resources), regulating (e.g. carbon sequestration, flood control, soil formation) and cultural (e.g. recreation, hunting and spiritual value of ecosystems).

Ecosystem process: The interactions among biotic and abiotic components of an ecosystem. This includes primary productivity, trophic transfers from plants to animals (e.g. herbivory, predation), nutrient cycling, decomposition, and heat transfer. Here we consider the term ecosystem process to be interchangeable with ecosystem function.

Insect herbivory: An ecosystem process involving the transfer of energy from plants to insects through feeding. Insect herbivores, also known as phytophages, are insects that feed at any life stage on any plant tissue or product including leaves, stems, bark, roots via any means such as chewing, sucking, or boring.

Non-outbreak species: Non-abundant or rare herbivore species that exist at low population densities, and often exhibit stable equilibrium population dynamics.

Outbreaking species: Herbivore species that exhibit unstable cyclic and irruptive population dynamics. As a result there are periods where these species may occur at high densities (outbreak phase) and low densities (latent phase).

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <http://dx.doi.org/10.1016/j.gecco.2015.05.006>.

References

- Aukema, B.H., Carroll, A.L., Zheng, Y., Zhu, J., Raffa, K.F., Moore, R.D., Stahl, K., Taylor, S.W., 2008. Movement of outbreak populations of mountain pine beetle: influences of spatiotemporal patterns and climate. *Ecography* 31, 348–358.
- Balvanera, P., Pfisterer, A.B., Buchmann, N., Nakashizuka, T., Raffaelli, D., Schmid, B., 2006. Quantifying the evidence for biodiversity effects on ecosystem functioning and services. *Ecol. Lett.* 9, 1146–1156.
- Baraza, E., Zamora, R., Hódar, J.A., 2010. Species-specific responses of tree saplings to herbivory in contrasting light environments: an experimental approach. *Ecoscience* 17, 156–165.
- Barbosa, V.S., Leal, I.R., Iannuzzi, L., Almeida-Cortez, J., 2005. Distribution pattern of herbivorous insects in a remnant of Brazilian Atlantic Forest. *Neotrop. Entomol.* 34, 701–711.
- Belovsky, G.E., Slade, J.B., 2000. Insect herbivory accelerates nutrient cycling and increases plant production. *Proc. Natl. Acad. Sci.* 97, 14412–14417.
- Belsky, A.J., 1986. Does herbivory benefit plants? A review of the evidence. *Am. Nat.* 127, 870–892.
- Bennett, E.M., Peterson, G.D., Gordon, L.J., 2009. Understanding relationships among multiple ecosystem services. *Ecol. Lett.* 12, 1394–1404.
- Biek, R., Real, L.A., 2010. The landscape genetics of infectious disease emergence and spread. *Mol. Ecol.* 19, 3515–3531.
- Blanchong, J.A., Samuel, M.D., Scribner, K.T., Weckworth, B.V., Langenberg, J.A., Filcek, K.B., 2008. Landscape genetics and the spatial distribution of chronic wasting disease. *Biol. Lett.* 4, 130–133.
- Bone, C., White, J.C., Wulder, M.A., Robertson, C., Nelson, T.A., 2013. Impact of forest fragmentation on patterns of mountain pine beetle-caused tree mortality. *Forests* 4, 279–295.
- Brandt, J., 1995. Forest insect- and disease-caused impacts to timber resources of west-central Canada: 1988–1992. Canadian Forest Service, Northern Forestry Centre, Northwest Information Report Vol. 341.
- Bryant, J.P., Chapin, F.S., Klein, D.R., 1983. Carbon nutrient balance of boreal plants in relation to vertebrate herbivory. *Oikos* 40, 357–368.
- Buhioff, G.J., Leuschner, W.A., 1978. Estimating psychological disutility from damaged forest stands. *Forest Sci.* 24, 424–428.
- Calabrese, J.M., Fagan, W.F., 2004. A comparison–shopper’s guide to connectivity metrics. *Front. Ecol. Environ.* 2, 529–536.
- Cappuccino, N., Lavertu, D., Bergeron, Y., Régnière, J., 1998. Spruce budworm impact, abundance and parasitism rate in a patchy landscape. *Oecologia* 114, 236–242.
- Chan, K.M.A., Shaw, M.R., Cameron, D.R., Underwood, E.C., Daily, G.C., 2006. Conservation planning for ecosystem services. *PLoS Biol.* 4, e379.
- Chapman, S.K., Hart, S.C., Cobb, N.S., Whitham, T.G., 2003. Insect herbivory increases litter quality and decomposition: an extension of the acceleration hypothesis. *Ecology* 84, 2867–2876.
- Clow, D.W., Rhoades, C., Briggs, J., Caldwell, M., Lewis Jr., W.M., 2011. Responses of soil and water chemistry to mountain pine beetle induced tree mortality in Grand County, Colorado, USA. *Appl. Geochem.* 26, S174–S178.
- Cooke, B.J., Lorenzetti, F., 2006. The dynamics of forest tent caterpillar outbreaks in Québec, Canada. *Forest Ecol. Manag.* 226, 110–121.
- Cooke, B.J., Roland, J., 2000. Spatial analysis of large-scale patterns of forest tent caterpillar outbreaks. *Ecoscience* 7, 410–422.

- Coulon, A., Cosson, J.F., Angibault, J.M., Cargnelutti, B., Galan, M., Morellet, N., Petit, E., Aulagnier, S., Hewison, A.J.M., 2004. Landscape connectivity influences gene flow in a roe deer population inhabiting a fragmented landscape: an individual-based approach. *Mol. Ecol.* 13, 2841–2850.
- Couture, J.J., Meehan, T.D., Kruger, E.L., Lindroth, R.L., 2015. Insect herbivory alters impact of atmospheric change on north temperate forests. *Nat. Plants* 1, 1–5. <http://dx.doi.org/10.1038/NPLANTS.2015.16>.
- Crowl, T.A., Crist, T.O., Parmenter, R.K., Belovsky, G., Lugo, A.E., 2008. The spread of invasive species and infectious disease as drivers of ecosystem change. *Front. Ecol. Environ.* 6, 238–246.
- Debinski, D.M., Holt, R.D., 2000. A survey and overview of habitat fragmentation experiments. *Conserv. Biol.* 14, 342–355.
- De La Vega, X., Grez, A.A., Simonetti, J.A., 2012. Is top-down control by predators driving insect abundance and herbivory rates in fragmented forests? *Austral Ecol.* 37, 836–844.
- DeLucia, E.H., 1999. Net primary production of a forest ecosystem with experimental CO₂ enrichment. *Science* 284, 1177–1179.
- Dyer, M.I., Turner, C.L., Seastedt, T.R., 1993. Herbivory and its consequences. *Ecol. Appl.* 3, 10–16.
- Dymond, C.C., Neilson, E.T., Stinson, G., Porter, K., MacLean, D.A., Gray, D.R., Campagna, M., Kurz, W.A., 2010. Future spruce budworm outbreak may create a carbon source in eastern Canadian forests. *Ecosystems* 13, 917–931.
- Ewers, R.M., Didham, R.K., 2006. Confounding factors in the detection of species responses to habitat fragmentation. *Biol. Rev.* 81, 117–142.
- Fischer, J., Lindenmayer, D.B., 2007. Landscape modification and habitat fragmentation: a synthesis. *Glob. Ecol. Biogeogr.* 16, 265–280.
- Foley, J.A., Defries, R., Asner, G.P., Barford, C., Bonan, G., Carpenter, S.R., Chapin, F.S., Coe, M.T., Daily, G.C., Gibbs, H.K., Helkowski, J.H., Holloway, T., Howard, E.A., Kucharik, C.J., Monfreda, C., Patz, J.A., Prentice, I.C., Ramankutty, N., Snyder, P.K., 2005. Global consequences of land use. *Science* 309, 570–574.
- Ford, M.A., Grace, J.B., 1998. Effects of vertebrate herbivores on soil processes, plant biomass, litter accumulation and soil elevation changes in a coastal marsh. *J. Ecol.* 86, 974–982.
- Frost, C., Hunter, M., 2004. Insect canopy herbivory and frass deposition affect soil nutrient dynamics and export in oak mesocosms. *Ecology* 85, 3335–3347.
- Frost, C.J., Hunter, M.D., 2007. Recycling of nitrogen in herbivore feces: plant recovery, herbivore assimilation, soil retention, and leaching losses. *Oecologia* 151, 42–53.
- Fuentealba, A., Bauce, É., 2012. Site factors and management influence short-term host resistance to spruce budworm, *Choristoneura fumiferana* (Clem.), in a species-specific manner. *Pest Manag. Sci.* 68, 245–253.
- Griffin, J.M., Turner, M.G., Simard, M., 2011. Nitrogen cycling following mountain pine beetle disturbance in lodgepole pine forests of Greater Yellowstone. *Forest Ecol. Manag.* 261, 1077–1089.
- Hennigar, C.R., Erdle, T.A., Gullison, J.J., MacLean, D.A., 2013. Re-examining wood supply in light of future spruce budworm outbreaks: a case study in New Brunswick. *For. Chron.* 89, 42–53.
- Hochwender, C.G., Sork, V.L., Marquis, R.J., 2003. Fitness consequences of herbivory on *Quercus alba*. *Am. Midl. Nat.* 150, 246–253.
- Hollenhorst, S.J., Brock, S.M., Freimund, W.A., Twery, M.J., 1993. Predicting the effects of gypsy moth on near-view aesthetic preferences and recreation appeal. *Forest Sci.* 39, 28–40.
- Hunter, M.D., 2001. Insect population dynamics meets ecosystem ecology: effects of herbivory on soil nutrient dynamics. *Agric For Entomol* 3, 77–84.
- Hunter, M.D., Linnen, C.R., Reynolds, B.C., 2003. Effects of endemic densities of canopy herbivores on nutrient dynamics along a gradient in elevation in the southern Appalachians. *Pedobiologia* 47, 231–244.
- James, P.M.A., Coltman, D.W., Murray, B.W., Hamelin, R.C., Sperling, F.A.H., 2011. Spatial genetic structure of a symbiotic beetle-fungal system: toward multi-taxa integrated landscape genetics. *PLoS One* 6, e25359.
- Jonsen, I.D., Fahrig, L., 1997. Response of generalist and specialist insect herbivores to landscape spatial structure. *Landscape Ecol.* 12, 185–197.
- Kareiva, P., 1990. Population-dynamics in spatially complex environments: theory and data. *Philos. Trans. Roy. Soc. B* 330, 175–190.
- Kenis, M., Auger-Rozenberg, M.-A., Roques, A., Timms, L., Péré, C., Cock, M.J.W., Settele, J., Augustin, S., Lopez-Vaamonde, C., 2009. Ecological effects of invasive alien insects. *Biol. Invasions* 11, 21–45.
- Kremen, C., Williams, N.M., Aizen, M.A., Gemmill-Herren, B., LeBuhn, G., Minckley, R., Packer, L., Potts, S.G., Roulston, T., Steffan-Dewenter, I., Vázquez, D.P., Winfree, R., Adams, L., Crone, E.E., Greenleaf, S.S., Keitt, T.H., Klein, A.-M., Regetz, J., Ricketts, T.H., 2007. Pollination and other ecosystem services produced by mobile organisms: a conceptual framework for the effects of land-use change. *Ecol. Lett.* 10, 299–314.
- Krieger, D.J., 2001. In: Kloepfer, Deanne (Ed.), *The Economic Value of Forest Ecosystem Services: A Review*. The Wilderness Society, Washington, DC, USA, pp. 1–31.
- Kruess, A., 2003. Effects of landscape structure and habitat type on a plant-herbivore-parasitoid community. *Ecography* 26, 283–290.
- Kruess, A., Tscharntke, T., 2000. Species richness and parasitism in a fragmented landscape: experiments and field studies with insects on *Vicia sepium*. *Oecologia* 122, 129–137.
- Kulman, H.M., 1971. Effects of insect defoliation on growth and mortality of trees. *Annu. Rev. Entomol.* 16, 289–324.
- Kurz, W.A., Dymond, C.C., Stinson, G., Rampley, G.J., Neilson, E.T., Carroll, A.L., Ebata, T., Safranyik, L., 2008. Mountain pine beetle and forest carbon feedback to climate change. *Nature* 452, 987–990.
- le Mellec, A., Gerold, G., Michalzik, B., 2011. Insect herbivory, organic matter deposition and effects on belowground organic matter fluxes in a central European oak forest. *Plant Soil* 342, 393–403.
- Leibold, M.A., Holyoak, M., Mouquet, N., Amarasekare, P., Chase, J.M., Hoopes, M.F., Holt, M.D., Shurin, J.B., Law, R., Tilman, D., Loreau, M., Gonzalez, A., 2004. The metacommunity concept: a framework for multi-scale community ecology. *Ecol. Lett.* 7, 601–613.
- Liebold, A.M., 2012. Forest pest management in a changing world. *Int. J. Pest Manag.* 58, 289–295.
- Loreau, M., Mouquet, N., Gonzalez, A., 2003. Biodiversity as spatial insurance in heterogeneous landscapes. *Proc. Natl. Acad. Sci.* 100, 12765–12770.
- Loreau, M., Naeem, S., Inchausti, P., et al., 2001. Biodiversity and ecosystem functioning: current knowledge and future challenges. *Science* 294, 804–808.
- Margosian, M.L., Garrett, K.A., Hutchinson, J.M.S., With, K.A., 2009. Connectivity of the American agricultural landscape: assessing the national risk of crop pest and disease spread. *BioScience* 59, 141–151.
- McNaughton, S.J., Oesterheld, M., Frank, D.A., Williams, K.J., 1989. Ecosystem-level patterns of primary productivity and herbivory in terrestrial habitats. *Nature* 341, 142–144.
- Metcalfe, D.B., Asner, G.P., Martin, R.E., et al., 2013. Herbivory makes major contributions to ecosystem carbon and nutrient cycling in tropical forests. *Ecol. Lett.* 17, 324–332.
- Mikkelsen, K.M., Bearup, L.A., Maxwell, R.M., Stednick, J.D., McCray, J.E., Sharp, J.O., 2013. Bark beetle infestation impacts on nutrient cycling, water quality and interdependent hydrological effects. *Biogeochemistry* 115, 1–21.
- Mitchell, M., Bennett, E.M., Gonzalez, A., 2013. Linking landscape connectivity and ecosystem service provision: current knowledge and research gaps. *Ecosystems* 16, 894–908.
- Müller, M., Job, H., 2009. Managing natural disturbance in protected areas: tourists' attitude towards the bark beetle in a German national park. *Biol. Cons.* 142, 375–383.
- Muzika, R.M., Liebold, A.M., 1999. Changes in radial increment of host and nonhost tree species with gypsy moth defoliation. *Can. J. For. Res.* 29, 1365–1373.
- Myers, J.H., Cory, J.S., 2013. Population cycles in forest Lepidoptera revisited. *Annu. Rev. Ecol. Syst.* 44, 565–592.
- Nicholson, E., Mace, G.M., Armsworth, P.R., Atkinson, G., Buckle, S., Clements, T., Ewers, R.M., Fa, J.E., Gardner, T.A., Gibbons, J., Grenyer, R., Metcalfe, R., Mourato, S., Muuls, M., Osborn, D., Reuman, D.C., Watson, C., Milner-Gulland, E.J., 2009. Priority research areas for ecosystem services in a changing world. *J. Appl. Ecol.* 46, 1139–1144.
- Paine, R.T., Tegner, M.J., Johnson, E.A., 1998. Compounded perturbations yield ecological surprises. *Ecosystems* 1, 535–545.
- Pedigo, L., Hutchins, S., Higley, L.G., 1986. Economic injury levels in theory and practice. *Annu. Rev. Entomol.* 31, 341–368.
- Pringle, C., 2003. What is hydrologic connectivity and why is it ecologically important? *Hydrol. Process.* 17, 2685–2689.
- Raffa, K.F., Aukema, B.H., Bentz, B.J., Carroll, A.L., Hicke, J.A., Turner, M.G., Romme, W.H., 2008. Cross-scale drivers of natural disturbances prone to anthropogenic amplification: the dynamics of bark beetle eruptions. *BioScience* 58, 501–517.

- Ribe, R.G., 1989. The aesthetics of forestry: what has empirical preference research taught us? *Environ. Manage.* 13, 55–74.
- Rinker, H.B., Lowman, M.D., 2004. Insect herbivory in tropical forests. In: Lowman, M., Rinker, H.B. (Eds.), *Forest Canopies*, second ed. Elsevier/Academic Press, San Diego, USA, pp. 359–386.
- Robert, L.-E., Kneeshaw, D., Sturtevant, B.R., 2012. Effects of forest management legacies on spruce budworm (*Choristoneura fumiferana*) outbreaks. *Can. J. For. Res.* 42, 463–475.
- Roland, J., 1993. Large-scale forest fragmentation increases the duration of tent caterpillar outbreak. *Oecologia* 93, 25–30.
- Roland, J., Taylor, P.D., 1997. Insect parasitoid species respond to forest structure at different spatial scales. *Nature* 386, 712–713.
- Safranyik, L., Carroll, A.L., Régnière, J., Langor, D.W., Riel, W.G., Shore, T.L., Peter, B., Cooke, B.J., Nealis, V.G., Taylor, S.W., 2010. Potential for range expansion of mountain pine beetle into the boreal forest of North America. *Can. Entomol.* 142, 415–442.
- Schowalter, T.D., 2012. Insect herbivore effects on forest ecosystem services. *J. Sustain. For.* 31, 518–536.
- Schowalter, T.D., Fonte, S.J., Geaghan, J., Wang, J., 2011. Effects of manipulated herbivore inputs on nutrient flux and decomposition in a tropical rainforest in Puerto Rico. *Oecologia* 167, 1141–1149.
- Schowalter, T., Hargrove, W., 1986. Herbivory in forested ecosystems. *Annu. Rev. Entomol.* 31, 177–196.
- Seidl, R., Rammer, W., Jäger, D., Lexer, M.J., 2008. Impact of bark beetle (*Ips typographus* L.) disturbance on timber production and carbon sequestration in different management strategies under climate change. *Forest Ecol. Manag.* 256, 209–220.
- Sheppard, S., Picard, P., 2006. Visual-quality impacts of forest pest activity at the landscape level: a synthesis of published knowledge and research needs. *Landsc. Urban Plann.* 77, 321–342.
- Souza, D.G., Santos, B.A., Wirth, R., Leal, I.R., Tabarelli, M., 2013. Community-level patterns of insect herbivory in a fragmented Atlantic forest landscape. *Environ. Entomol.* 42, 430–437.
- Stadler, B., Solinger, S., Michalzik, B., 2001. Insect herbivores and the nutrient flow from the canopy to the soil in coniferous and deciduous forests. *Oecologia* 126, 104–113.
- Taylor, P.D., Fahrig, L., Henein, K., Merriam, G., 1993. Connectivity is a vital element of landscape structure. *Oikos* 68, 571–573.
- Terborgh, J., 2001. Ecological meltdown in predator-free forest fragments. *Science* 294, 1923–1926.
- Thies, C., Steffan-Dewenter, I., Tscharrntke, T., 2003. Effects of landscape context on herbivory and parasitism at different spatial scales. *Oikos* 101, 18–25.
- Thomas, J.A., Bourn, N.A., Clarke, R.T., Stewart, K.E., Simcox, D.J., Pearman, D.S., Curtis, R., Goodger, B., 2001. The quality and isolation of habitat patches both determine where butterflies persist in fragmented landscapes. *Proc. Biol. Sci.* 268, 1791–1796.
- Tilman, D., Balzer, C., Hill, J., Befort, B.L., 2011. Global food demand and the sustainable intensification of agriculture. *Proc. Natl. Acad. Sci.* 108, 20260–20264.
- Tscharrntke, T., Brandl, R., 2004. Plant–insect interactions in fragmented landscapes. *Annu. Rev. Entomol.* 49, 405–430.
- Tscharrntke, T., Dewenter, I.S., Kruess, A., 2002. Characteristics of insect populations on habitat fragments: a mini review. *Ecol. Res.* 17, 229–239.
- Uunila, G.A.P., Guy, B., Pike, R., 2006. Hydrologic effects of mountain pine beetle in the interior pine forests of British Columbia: key questions and current knowledge. *JEM* 7, 37–39.
- van der Wal, R., Bardgett, R.D., Harrison, K.A., Stien, A., 2004. Vertebrate herbivores and ecosystem control: cascading effects of faeces on tundra ecosystems. *Ecography* 27, 242–252.
- Vanderzee, D., 1990. The complex relationship between landscape and recreation. *Landsc. Ecol.* 4, 225–236.
- Vitousek, P.M., Mooney, H.A., Lubchenco, J., Melillo, J.M., 1997. Human domination of Earth's ecosystems. *Science* 277, 494–499.
- Wallner, W.E., 1987. Factors affecting insect population dynamics: differences between outbreak and non-outbreak species. *Annu. Rev. Entomol.* 32, 317–330.
- Wood, D.M., Parry, D., Yanai, R.D., Pitel, N.E., 2010. Forest fragmentation and duration of forest tent caterpillar (*Malacosoma disstria* Hübner) outbreaks in northern hardwood forests. *Forest Ecol. Manag.* 260, 1193–1197.
- Yang, L.H., Gratton, C., 2014. Insects as drivers of ecosystem processes. *Curr. Opin. Insect Sci.* 2, 26–32.
- Ziter, C., Bennett, E.M., Gonzalez, A., 2013. Functional diversity and management mediate aboveground carbon stocks in small forest fragments. *Ecosphere* 4, art85.